

APPLIED RESEARCH

Hybrid Data Lake Entities Localization Optimization Based on Complex Networks

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ABSTRACT Data Lakes are a relevant analytics architecture, with current implementations in hybrid and multi-cloud environments presenting governance challenges in cost management and data localization. This work proposes the “Entity Migration Recommender System,” an engine that ingests Data Lake metadata to create optimized complex network digraphs with recommended entity localities. The tool demonstrated a 16.2% overall cost reduction by optimizing data entity placement across cloud and on-premise infrastructure. This method leverages complex network metrics to address the critical dimension of data interoperation costs in distributed analytics environments.

INDEX TERMS Cloud computing, complex networks, data analytics, data lakes.

I. INTRODUCTION

Data Lakes emerged as a data architecture related to Big Data scenarios, based on ingesting large amounts of data in raw format [1], and creating a logical centralized data structure to serve as input for Data Science, increasing data usability and agility [2].

Based on an increasing demand for infrastructure resources, intending to attend a massive growth in data ingestion and processing in the Data Lake, enterprise organizations adopted strategies for expanding self owned, on-premise capacities, taking benefit of cloud computing elastic capabilities and defining new practices of Data Lakes deployed in cloud computing providers, referred to as Cloud Data Lakes. In a competitive market, with multiple cloud providers offering distinguished services and competing on price and performance, enterprise organizations considered not only the design of hybrid clouds Data Lakes, with on-premise and cloud deployments, but also the potential of multi cloud scenarios [3].

In a storage level perspective, a Hybrid Data Lake could consider on-premise storage on top of distributed file systems, but also Cloud Data Lake capabilities in object storages such as Amazon S3, Azure Data Lake Storage or Google

Cloud Storage [4], enabling the potential of federated and heterogeneous data operations, like aggregations, filters and projections in a standardized SQL language for data users operations [5].

Considering the Data Governance perspective, Data Lakes evolved to Curated structures composed of zones [6], process and metadata [7], with a proposition of a centralized Data Lake Catalog [8], having business and technical metadata tags, such as schemas and data localization. This extends the definition of Data Lakes as physical centralized data structures to logical centralized data structures with metadata.

With the infrastructure and platform capabilities and having a centralized catalog that enables the data discovery in distributed environments, enterprise organizations could operate the distributed lake. However the cost involved in the data interoperation between distributed entities impacts governance and operations. The existence of massive data entities with potential data gravity [9] and the cost differentiators between cloud providers, brought challenges on defining optimized localizations for data entities, creating the opportunities for methods addressing the best localizational strategy for data.

This article applies the concept of complex networks in Data Lake entities localization strategy and is structured as follows: Chapter I is the Introduction, Chapter II defines the costs variables in hybrid and multi cloud environments,

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Chapter III cover the applicability of Complex Networks in Data Lake, Chapter IV presents the “Entity Migration Recommender System” and the analysis and Chapter V contains the conclusions and Future works.

II. BACKGROUND AND RELATED WORKS

A. COSTS VARIABLES IN HYBRID AND MULTI CLOUD INFRASTRUCTURES

Cloud computing offers new technical capabilities when compared to on-premise structures, being possible that organizations select between a composition involving on-premise environment and cloud providers, being possible to have hybrid and multi-cloud scenarios [3].

In a technology economy perspective, Cloud Computing tends to reduce the Capital Expenditure (CAPEX), having the Operating Expenses based on resources utilized [10]. However, organizations may opt to maintain on-premise infrastructure, either due to legal requirements or to utilize resources already acquired.

Operational costs in On-premise Data Lakes involves infrastructure costs, engineering resources and experts. While in Cloud Computing it is required to have engineering resources and experts with cloud costs dimensions. Analyzing the three biggest public cloud providers (Amazon Web Services, Microsoft Azure and Google Cloud) [3], it was possible to collect three dimensions related to Data Lake environment:

- **Storage:** costs related to data storage, having potential variation according to related technology and archiving characteristics. Costs in storage layers are related to amount of data at rest and can have related operations fees based on the selected storage strategy, like standard and archiving layers [11];
- **Compute:** costs related to data processing capabilities, having potential variations according to technology. Compute resources are associated with processing capacity, being able to be Iaas (Infrastructure as a Service), Paas (Platform as a Service) or SaaS (Software as a Service) [12], with each strategy pursuing specific rates.
- **Egress charges:** charges from cloud providers for moving data from the cloud storage layer [13]. Costs between cloud internal moving differs from moving to external cloud providers.

Additional Cloud managed services might also have usage fees and the assessed cloud providers offered minimum usage rates with costs as incentives for initial consumption. In cloud, compute resource management, associated with correct storage and egress strategy are relevant practices to minimize the operational costs [14].

B. DATA LAKE AND COMPLEX NETWORKS

The application of complex networks for analytics data entities intends to extract network metrics and also provide visual knowledge patterns useful for governance procedures [15].

This practice is extensible to Data Lake environments, a centralized structure with massive organization of data entities with multiple usage purposes [5].

Farrugia et.al. [16] proposed the application of complex networks to analyze the detection of clusters and the scale-free network properties intending to define the most relevant and influential data assets, communities and proving the feasibility of complex network metrics and properties to enhance the Data Lake management.

The usage of common Data Lake metadata, such as lineage, tags and ETL jobs logs, available in the Spark proposed architecture, denotes a viable data source for the construction and application of complex networks algorithms.

Based on the proven feasibility of complex networks in Data Lake metadata [16], this work proposes costs calculation methods based on degree metrics and vertex properties in a hybrid Data Lake environment.

III. EXPERIMENT AND COMPUTATIONAL DETAILS

A. DATA LAKE COMPLEX NETWORK COST CALCULATION

The cloud costs for analytics is defined in the experiment with 3 major dimensions: storage, compute and egress. To enable a comparison between cloud and on-premise environments in the network, it is considered a fixed cost per each dimension, not assuming variations in cloud managed services fees.

A representation of the interrelation in a Data Lake is given by a Graph $G(V,A)$ with n vertices and m edges. The graph is directed, acyclic, with the edges containing the volume of data transferred between tables and the elapsed processing time in seconds. The vertices in the modelling also contain properties, having the volume of data stored in the entity and location, being Cloud or On-Premise (OP).

The figure 1 illustrates a sample data entity graph with its related metadata in nodes and edges.



FIGURE 1. Sample entities graph.

Figure 1 illustrates a group of 5 data entities, where t1 is the source of t2,t3 and t4 and t2 also originates the t5. Nodes

and edges have properties related to data storage, volume of egress data and job elapsed time.

The following function represents the individual cost of a data entity within a Data Lake:

$$C_i = \alpha_i \cdot v_i + \sum_1^n s_{ji} \cdot \beta_{ji} + \sum_1^n p_{ji} \cdot \gamma_{ji} \quad (1)$$

where α_i refers to the storage rate of the for entity i (in Gigabytes), v_i refers to the total data volume of entity i (in Gigabytes), s_{ji} the volume of data trafficked between the predecessor edges to entity i , β_{ji} is the current data egress rate from source entity j to destination entity i , p_{ji} the elapsed time to process entity j to entity i (seconds) and γ_{ji} the data processing rate (seconds).

The total cost of a Data Lake, for a set of n entities, is given by:

$$C_t = \sum_1^n C_i \quad (2)$$

where C_i refers to the individual cost of the entity i .

The function used to categorize an entity as a candidate for migration is defined by the function:

$$M_i = C_i / C'_i \quad (3)$$

where C_i is the individual calculation of entity i in its current state, not migrated, based on the function (1), but considering the out degree on edges:

$$C'_i = \alpha \cdot v_i + \sum_1^n s_{ji} \cdot \beta_{ji} + \sum_1^n s_{ik} \cdot \beta_{ik} + \sum_1^n p_{ij} \cdot \gamma_{ji} \quad (4)$$

with referring to the storage rate of the entity's current environment (in Gigabytes), v_i the total data volume of entity i (in Gigabytes), s_{ji} the volume of data trafficked between the edges that give rise to entity i , β_{ji} the current data egress rate from source entity j to destination entity i to be migrated, s_{ik} the volume of data trafficked between the source i and the target k , β_{ik} the current data egress rate from source entity i to destination entity k , p_{ij} the time taken to process entity i to resulting entity j (in seconds), γ_{ji} the data processing rate (in seconds) in the new migration environment.

C'_i is based on the expression 4, but considering the entity as migrated:

$$C'_i = \alpha' \cdot v_i + \sum_1^n s_{ji} \cdot \beta_{ji} + \sum_1^n s_{ik} \cdot \beta_{ik} + \sum_1^n p_{ij} \cdot \gamma'_{ij} \quad (5)$$

with α' referring to the storage rate of the entity's target environment (in Gigabytes), v_i the total data volume of entity i (in Gigabytes), s_{ji} the volume of data trafficked between the edges that give rise to entity i , β_{ji} the current data egress rate from source entity j to destination entity i to be migrated, s_{ik} the volume of data trafficked between the source i and the target k , β'_{ik} the current data egress rate from migrated entity i to destination entity k , p_{ij} the time taken to process entity i

to resulting entity j (in seconds), γ'_{ij} the data processing rate (in seconds) in the new migration environment.

Based on the categorization function denoted by M_i , the resulting value of the function is directly proportional to the likelihood of the entity to be migrated, i.e., $M_i > 1$.

B. ENTITY MIGRATION RECOMMENDER SYSTEM

The experiment considered the execution of the Entity Migration Recommender System using public data and synthetic data based on data manipulation from the public data.

The Data Lake was constructed based on initial public Brazilian health care data, from official government entities based on DataSus Tabnet [17]. As processing infrastructure, Apache Spark was selected due to its popularity in the Big Data environment and performance oriented structure [18].

The raw entities from DataSus Tabnet data sources were defined as “base” entities. As the amount of base entities were insufficient to reproduce a Data Lake with complex properties as suggested by references [16], new simulated entities were generated using Spark SQL “CREATE AS SELECT” statements, these were named as “Simulated Entities”.

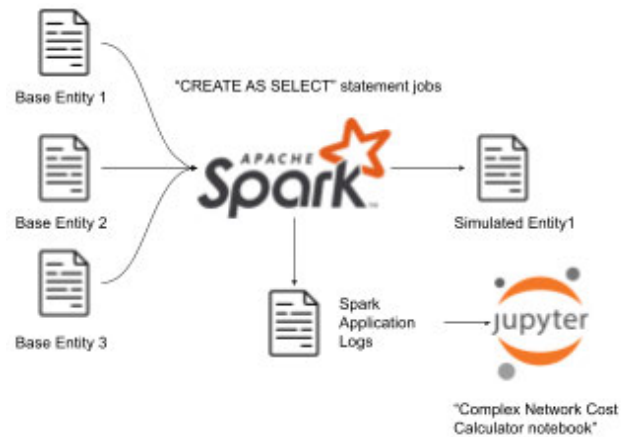


FIGURE 2. Entities creation and metadata generation architecture.

The figure 2 illustrates the “Simulated Entities” creation and logs ingestion. The Spark Logs were generated based on the creation of the “Simulated Entities” with usage of SparkSQL in “Create as Select” statements. These logs were ingested by the “Complex Network Cost Calculator Notebook”, a python notebook which creates a network digraph object, having into vertices the name, size and the location of the data entity and into edges the elapsed time used to generate the data entity. The referred location into the vertex was based using a random function, having as possible values “on-premise” and “cloud” reflecting the hybrid cloud strategy.

Following the properties observed by Farrugia et.al. [16] into the complex network observed into a production Data Lake, 30% of the base entities were selected to have 70% of the amount of edges, being part of the Spark SQL statement, reproducing a scale-free network characteristic. This

proportion was assumed on the definition of quantities in simulated entities and edges/relations between entities.

TABLE 1. Data Sources and generated components.

Objects	Count
Base entities	34
Simulated entities	70
Edges/Relations	103

As standard values for storage, cost and egress rates, it was defined as configurable infrastructure rates, in the format of variables for cloud and on-premise environments.

The following digraph represents the initial digraph, based on base and simulated entities and the relations.

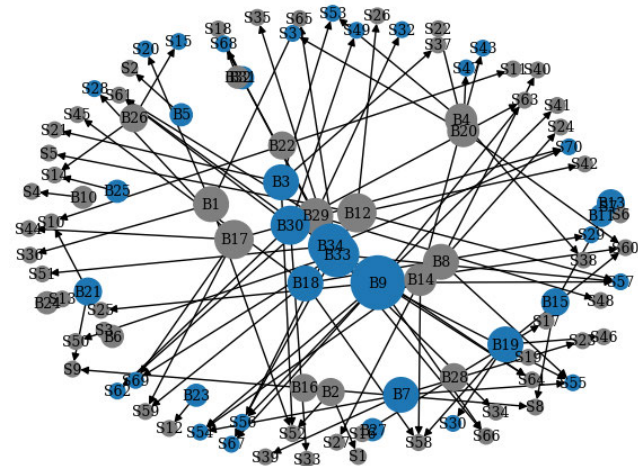


FIGURE 3. Initial digraph based on-premise and cloud entities.

In the representation, a created digraph, where the vertex size is proportional to the out-degree value, being on-premise entities represented in gray, and cloud entities represented in blue. The simulation considered 38 entities in cloud infrastructure and 66 entities in on-premise infrastructure.

Calculating the sum of base entities, simulated entities and their relationships the total amount is 88916.61 units considering the following rates proportions assumed as infrastructure representation.

The proportions were illustrative considering a hybrid environment. Defining a baseline for on-premise versus cloud costs is a challenge, requiring consideration of Capex, long-term TCO, licences and local taxes [19]. For the experiment, the on-premise storage costs were 20% higher for storage and 30% higher for computation resources, having a zero rate for egress. In addition, the cloud variable assumed a 0.08 cost rate for egress, considering data movement from the cloud provider to the on-premise infrastructure.

TABLE 2. Infrastructure cost factors in experiment.

Variable	Value (illustrative unit)
onprem_storage_cost	1.2
onprem_egress_cost	0.0
onprem_compute_cost	1.3
cloud_storage_cost	1
cloud_egress_cost	0.08
cloud_compute_cost	1

IV. RESULTS AND DISCUSSION

A. MIGRATION RECOMMENDATIONS

Having the initial digraph structure with the applied costs variables, the “Entity Migration Recommender System” is capable of navigating in all vertices, applying the migration categorization function represented by M_i and creating a flag indicating if the data entity must be migrated.

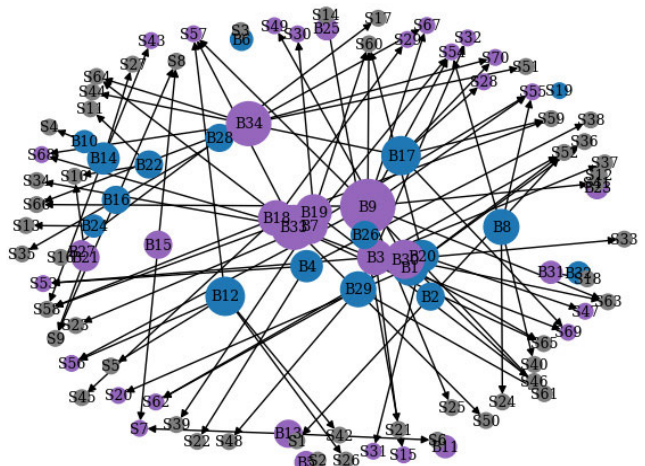


FIGURE 4. Resulting digraph of a post entity migration recommender system execution.

The resulting digraph represents the same 70 vertices and edges digraph from initial simulation, however considering proposed data movement to optimize costs. The gray vertices represent on-premise entities that must persist on-premise, purple represents entities situated on-premise that might be moved to cloud, blue represents cloud entities that must persist in the current cloud provider and brown represents entities situated in cloud that might be moved to on-premise.

The total cost of the new migrated digraph is 76514.29, observing a 16.2% cost reduction compared to the initial digraph. This method was executed iteratively, observing that the resulting digraph did not select new migration candidate vertices and without modifying the total cost.

B. ANALYSIS OF INFRASTRUCTURE WEIGHTS

Analyzing the resulting migration digraph, it is possible to verify the non-existence of brown vertices, indicating the financial advantage of the cloud environment for the simulated storage and computing fees. An important factor observed is the data egress, indicating that the product obtained of the volume of egress data and egress fee did not overcome the storage and compute costs.

The analyzed break-even for the equation is based on the sum of differences of storage, compute versus the related egress costs.

C. ANALYSIS OF INTERACTION FACTOR

The Data Lake metadata digraph represents a dynamic structure, where changing a locality of an entity can impact the cost on entities not initially recommended for migration, due to the existence of the egress as a calculation factor.

This behavior is confirmed during the iterative execution of the “Entity Migration Recommender System”.

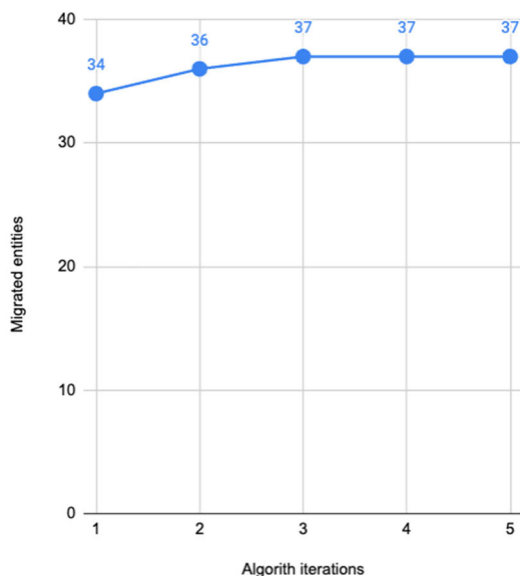


FIGURE 5. Algorithm iteration x migrated entities chart.

Analyzing the chart it is possible to verify that the first execution performed the suggested migration of 34 entities, evolving to 36 entities in the second iteration and stabilizing in 37 entities in the third iteration.

A potential inference is that the number of required iterations is directly proportional with the weight of egress rate.

V. CONCLUSION

The “Entity Migration Recommender System” successfully identified candidate data entities for migration, leading to a significant 16.2% overall cost reduction by optimizing their location within a hybrid Data Lake environment. The execution converged to an optimal point after three algorithm executions.

The algorithm’s decision-making process was oriented by illustrative infrastructure costs proportions, where cloud offered lower rates for storage and compute fees, as evidenced by the absence of recommended migrations from cloud to on-premise. While data egress costs were considered, their impact did not outweigh the benefits derived from optimized storage and compute resource allocation.

This algorithm can also support the decision between two cloud providers, where prices for infrastructure resources can be deterministic based on cloud providers products references.

This work complements the existing knowledge of complex networks into Data Lake governance practices [16], enabling organizations to evaluate and control costs on hybrid infrastructure.

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